

Pallav Gupta

Senior Software Engineer · Platform Engineer · Cloud-Native & Distributed Systems

+91 7415172691 | pallforwork2@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Platform and distributed systems engineer with 6+ years of end-to-end ownership from architecture to production across telecom, AI, and enterprise domains. At Nokia, led design and delivery of a cloud-native AI positioning platform and an application lifecycle management system, both deployed on Kubernetes at the edge. Deep expertise in event-driven microservices (FastAPI, Node.js, MQTT, ZeroMQ), container orchestration (K3S, Helm, ArgoCD), and polyglot persistence (PostgreSQL, MongoDB, ClickHouse). Comfortable operating at the intersection of systems design, DevOps, and product delivery.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+, Shell Scripting), Java

Backend: FastAPI, Flask, Node.js, Express.js, NestJS, Spring Boot

Frontend: ReactJS, HTML5, CSS3

Cloud & DevOps: Docker, Kubernetes, K3S, Helm, ArgoCD, Microsoft Azure, Nginx, Apache2, GitLab CI/CD

Messaging: MQTT, ZeroMQ, HTTP/2, REST APIs, JWT Authentication

Databases: PostgreSQL, MariaDB, MongoDB, ClickHouse, Oracle DB

Concepts: Microservices Architecture, System Design, Distributed Systems, Edge Computing, CI/CD, GitOps, Cloud-Native Applications

PROFESSIONAL EXPERIENCE

HCL Software · Bengaluru, IN

April 2026 – Present ★ **Current**

Senior Software Engineer 2

Augmented Network Automation

Stack: Python, FastAPI, ClickHouse, MQTT, ZeroMQ, Docker, Helm, Kubernetes (K3S), HTTP/2

- ▶ Owned migration of source code repositories from Bitbucket to GitLab for 6 teams of 75+ engineers, maintaining zero disruption to CI/CD pipelines and release cadence.
- ▶ Led database modernization initiative, migrating persistence layer from ClickHouse to MongoDB and integrating new ORM to improve query flexibility and maintainability.
- ▶ Designed and deployed containerized cloud-native services on Kubernetes (K3S) using Helm charts, improving deployment repeatability across environments.

Nokia · Bengaluru, IN

April 2022 – April 2026

Platform Lead

Context-Aware AI Positioning Platform

Stack: Python, FastAPI, PostgreSQL, MQTT, ZeroMQ, Docker, Helm, Kubernetes (K3S), HTTP/2

- ▶ Led end-to-end design and delivery of a cloud-native platform orchestrating multiple AI models for real-time context-aware positioning, sole architect across the full lifecycle from whiteboard to production.
- ▶ Architected event-driven communication layer using MQTT and ZeroMQ, enabling low-latency interaction between distributed platform components at the edge.
- ▶ Reduced API latency by 40% and optimised system throughput to support 4× more camera streams per edge device, directly improving deployment economics for end customers.
- ▶ Built containerised deployment workflows using Docker, Helm, and K3S for edge and IoT environments; maintained high availability across distributed nodes.

Application Management Platform

Stack: Node.js, Express.js, Spring Boot, Python, ReactJS, PostgreSQL, MQTT, ArgoCD, Kubernetes, Helm

- ▶ Led development of a full-stack platform for application lifecycle management, handling deployment, monitoring, and operational management of distributed applications at enterprise scale.
- ▶ Built backend services in Node.js, Spring Boot, and Python; developed a ReactJS management console providing a unified operator interface.
- ▶ Integrated Kubernetes, Helm, and ArgoCD into a GitOps-based deployment pipeline, reducing manual deployment steps and improving release consistency.

Infosys · Bengaluru, IN

Dec 2021 – April 2022

Associate Consultant

Common Authentication System

Stack: ReactJS, Node.js, Express.js, NestJS, MongoDB, Apache, Shell Scripting

- ▶ Developed a centralised authentication platform designed as a reusable plugin for enterprise applications, reducing onboarding effort for new integrating services.
- ▶ Migrated core backend services from Express.js to NestJS, improving code modularity and long-term maintainability.
- ▶ Configured Apache for routing, load balancing, and secure SSL termination across environments.

Tata Consultancy Services · Gandhinagar, IN

July 2019 – Dec 2021

Systems Engineer

Telecom Solutions & Cloud Infrastructure

Stack: Node.js, ReactJS, Spring Boot, Azure, PostgreSQL, MariaDB, Oracle DB, Shell Scripting

- ▶ Built and maintained enterprise backend services and APIs for telecom customers across India and Europe using Java Spring Boot and Node.js.
- ▶ Developed customer-facing web applications and reusable ReactJS component libraries used across multiple products.
- ▶ Managed deployment, scaling, and operational support of application and database infrastructure on Microsoft Azure, covering PostgreSQL, MariaDB, and Oracle DB.
- ▶ Established secure Express Route VPN connectivity between Azure-hosted services and customer on-premises environments.

EDUCATION

Madhav Institute of Technology and Science · Gwalior, IN

2015 – 2019

Bachelor of Engineering, Mechanical Engineering · CGPA 7.1

Transitioned to software engineering through professional development and self-study; joined TCS as Systems Engineer upon graduation in 2019.